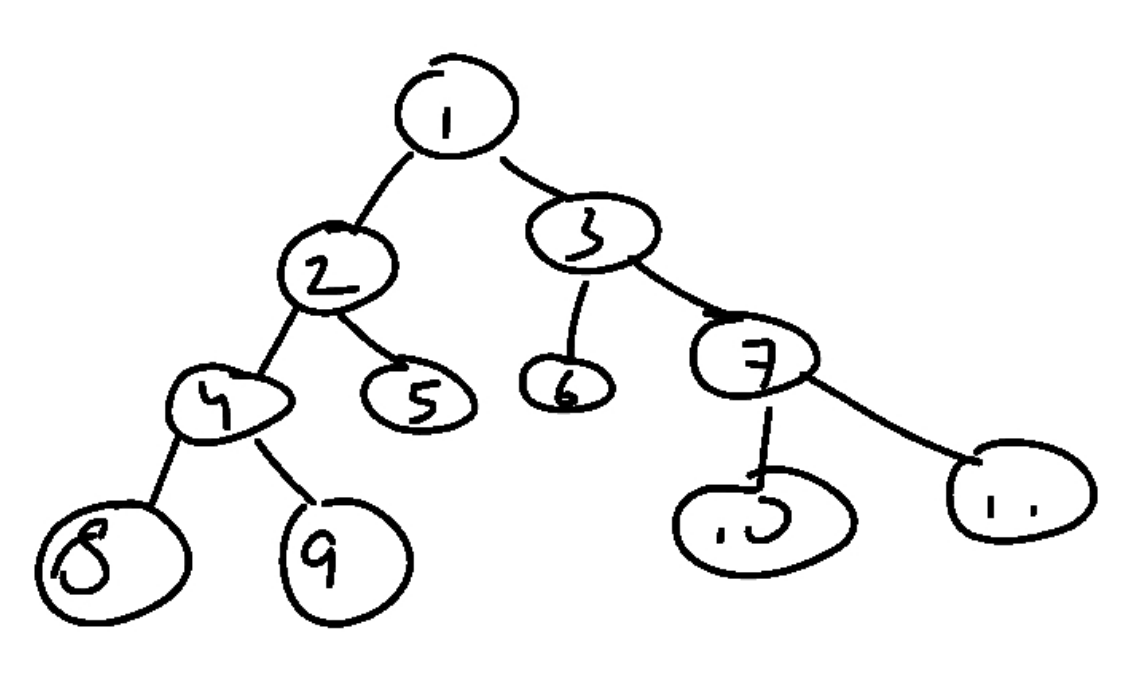
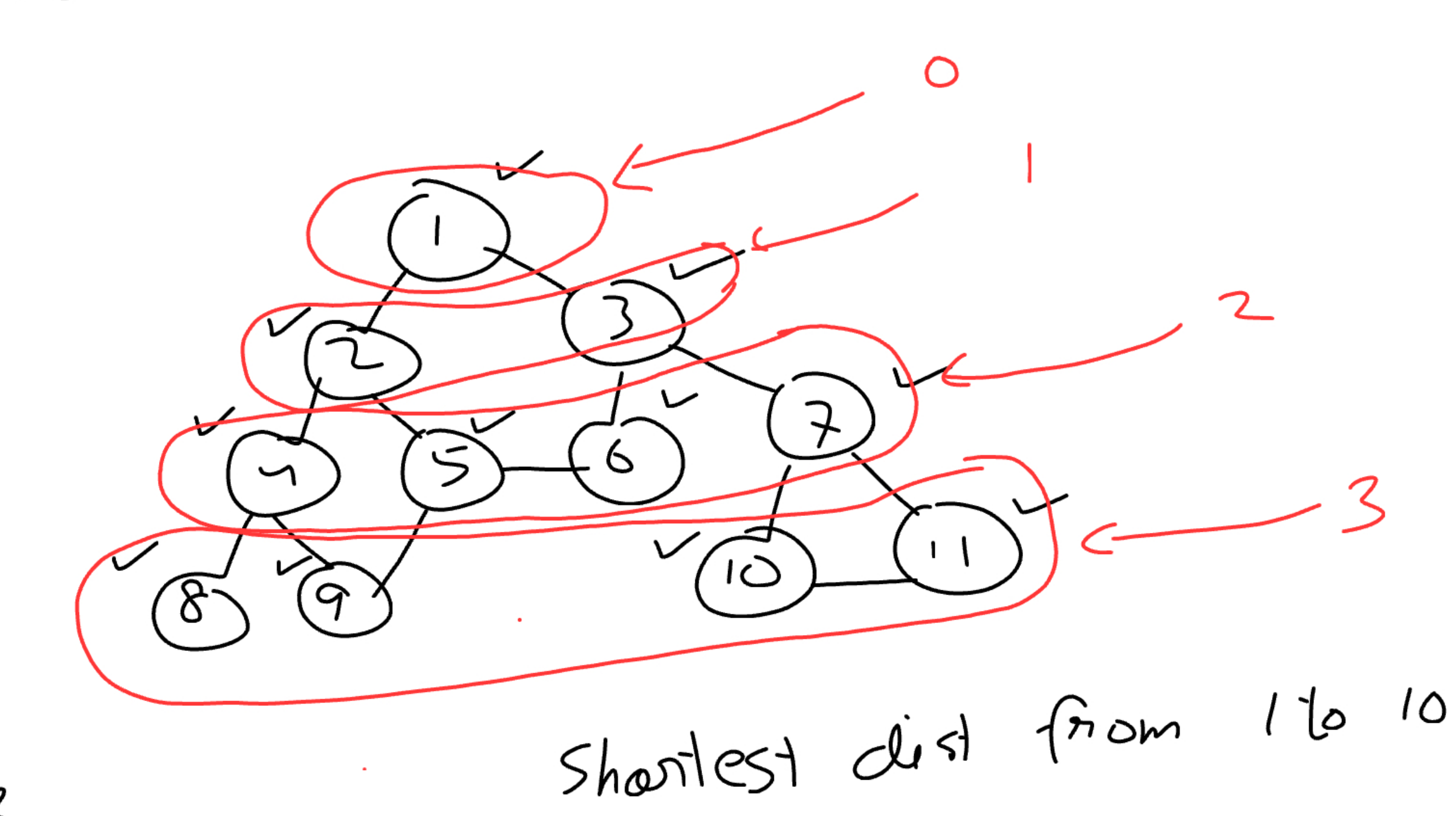

1
2
3
4
5
6
7
8
9
10
11

1
2 3
4, 5, 6, 7
8, 9, 10, 11

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

level order traversal on BT → BFS.


0
1
2
3
1
2
3
4
5
6
7
8
9
10
11

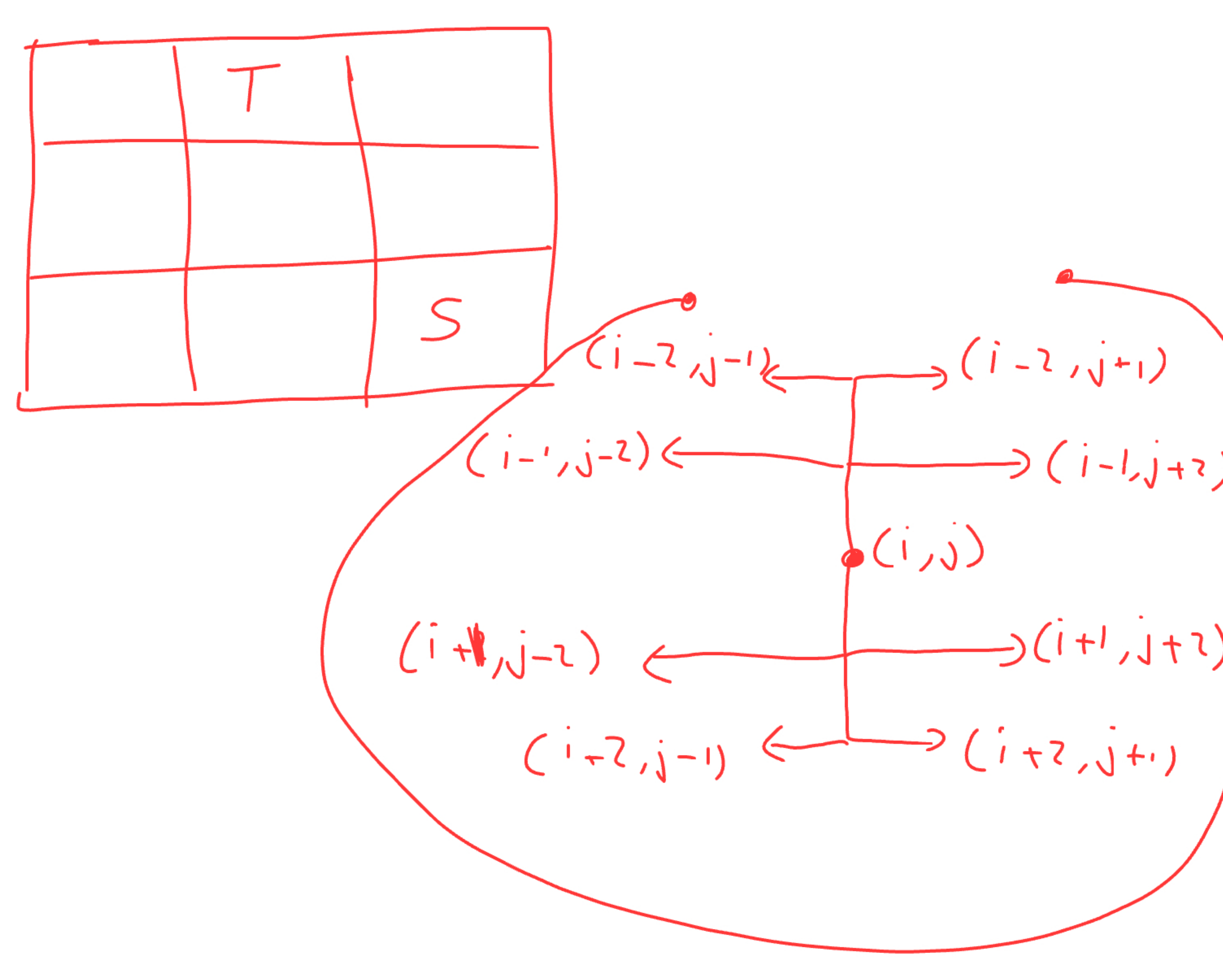
Shortest dist from 1 to 10.

1 → 2, 3
2 → 1, 4, 5
3 → 1, 6, 7
4 → 2, 8, 9
5 → 2, 6, 9
6 → 3, 5
7 → 3, 10, 11
8 → 4
9 → 4, 5
10 → 7, 11
11 → 7, 10

sz = ~~1~~ ~~2~~ 4

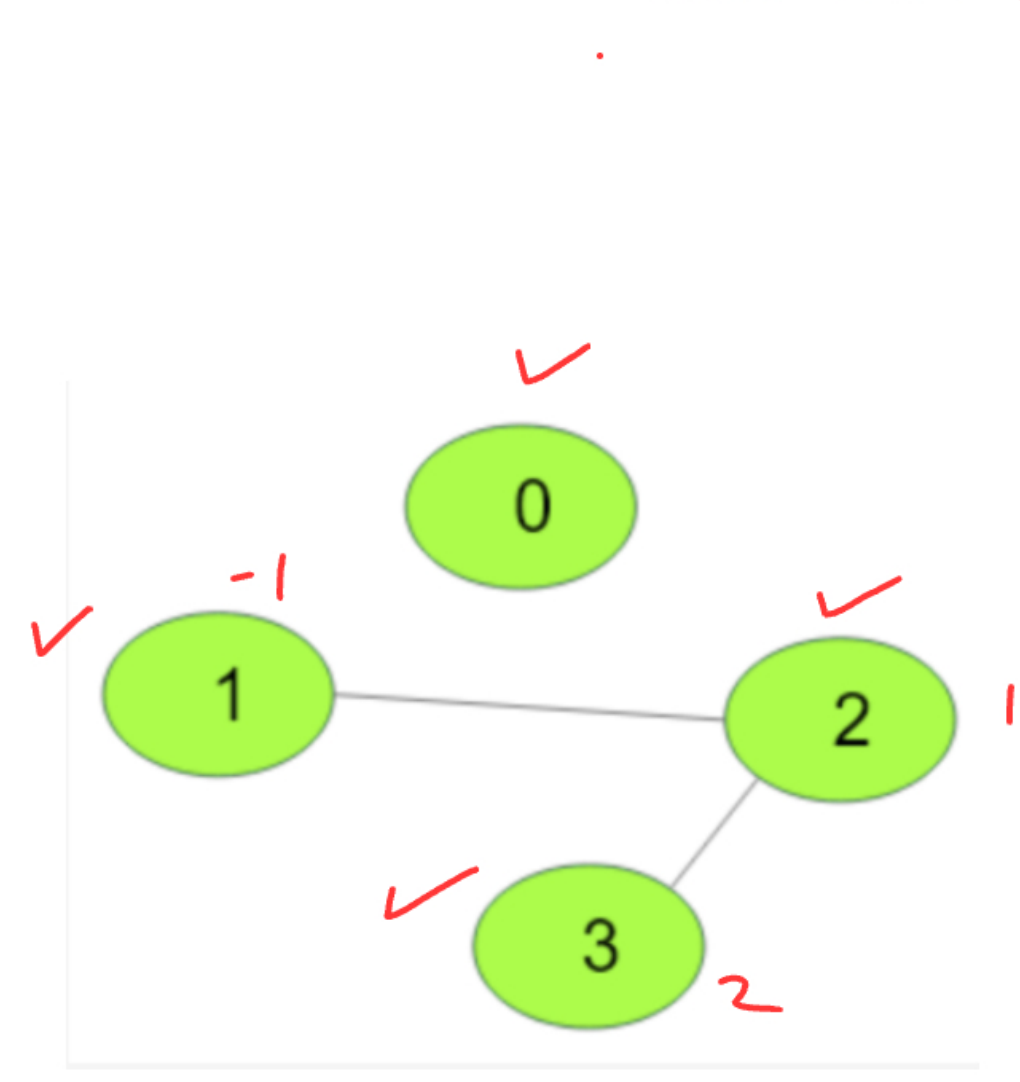
Q: ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~5~~, ~~6~~, ~~7~~, 8, ~~9~~, 10, ~~11~~

SSSP Algo → Single Source Shortest Path Algo.


T
S
(i-2, j-1)
(i-2, j+1)
(i-1, j-2)
(i-1, j+2)
(i, j)
(i+1, j-2)
(i+1, j+2)
(i+2, j-1)
(i+2, j+1)

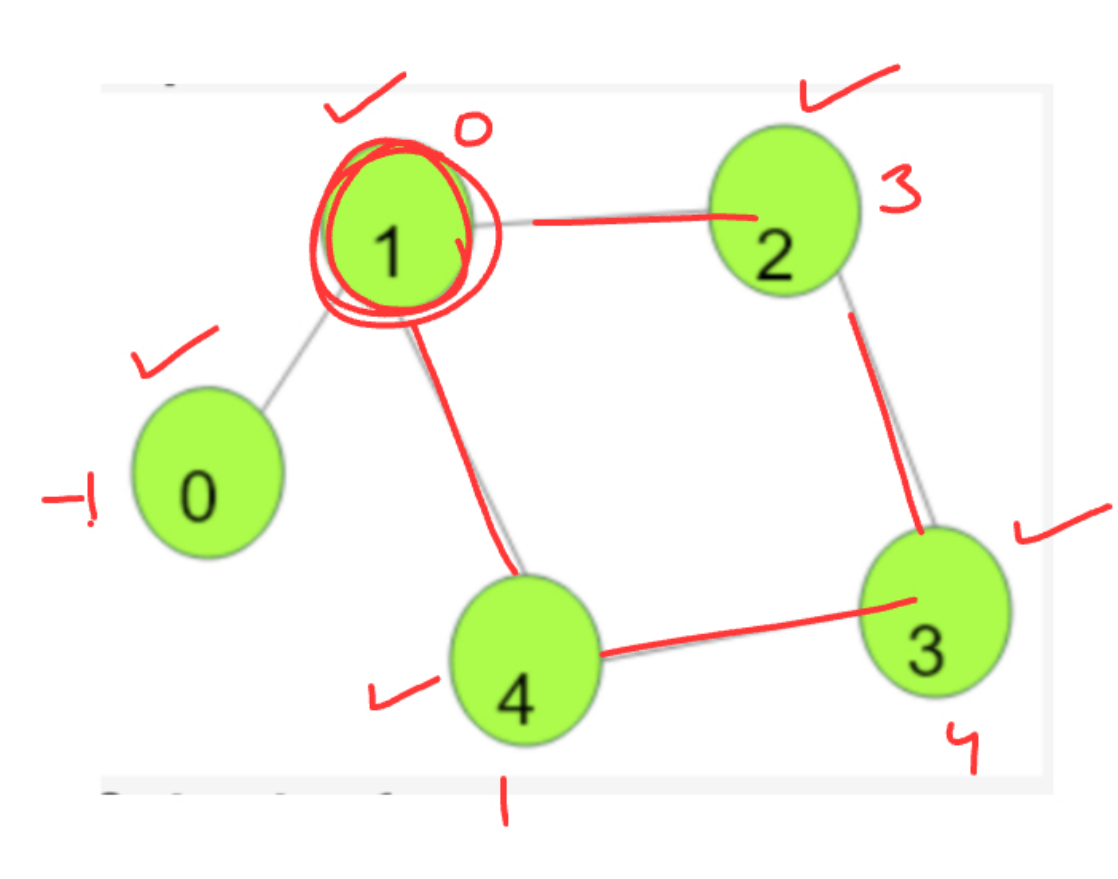
TODO.

int shortestDis (adjList, src, target)

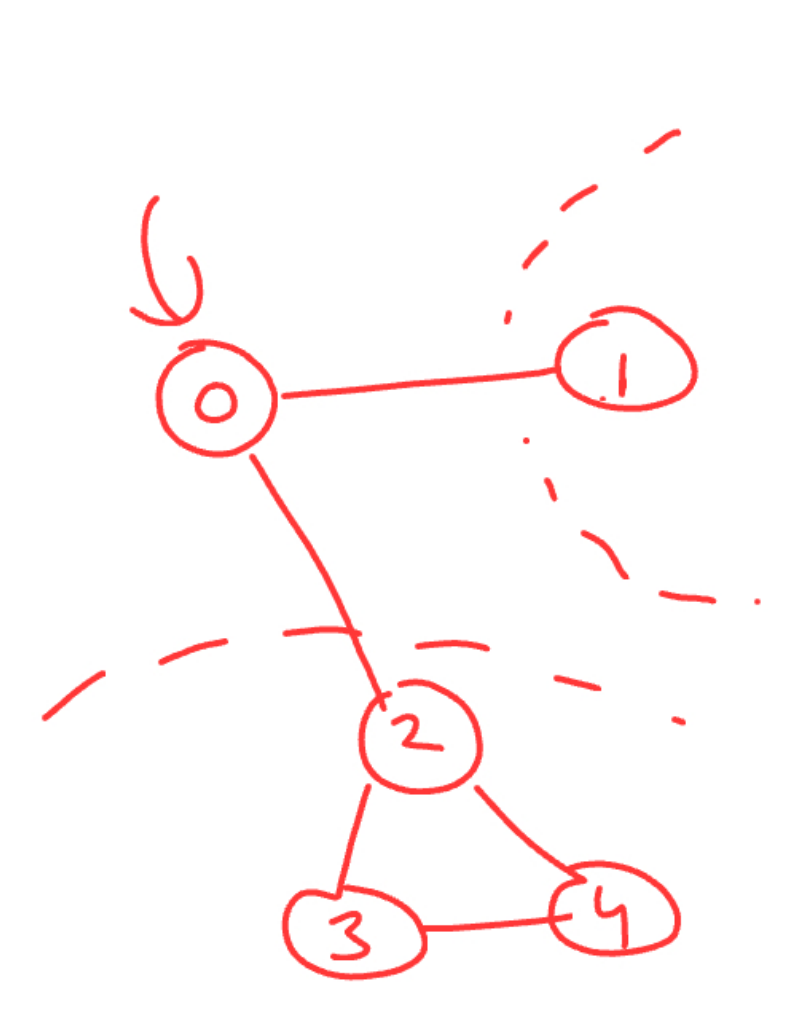


0 →
1 → 2
2 → 1, 3
3 → 2

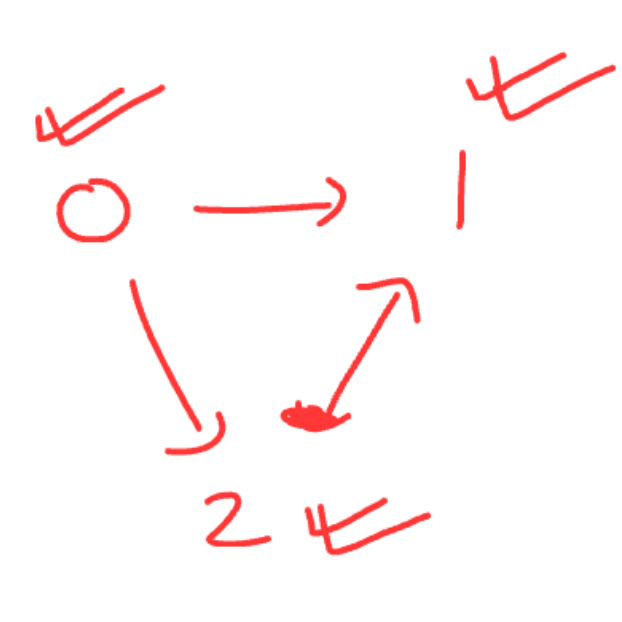
0 → 1
1 → 0, 4, 2
2 → 1, 3
3 → 2, 4
4 → 1, 3


-1
0
1
0
2
3
4
3
1
4

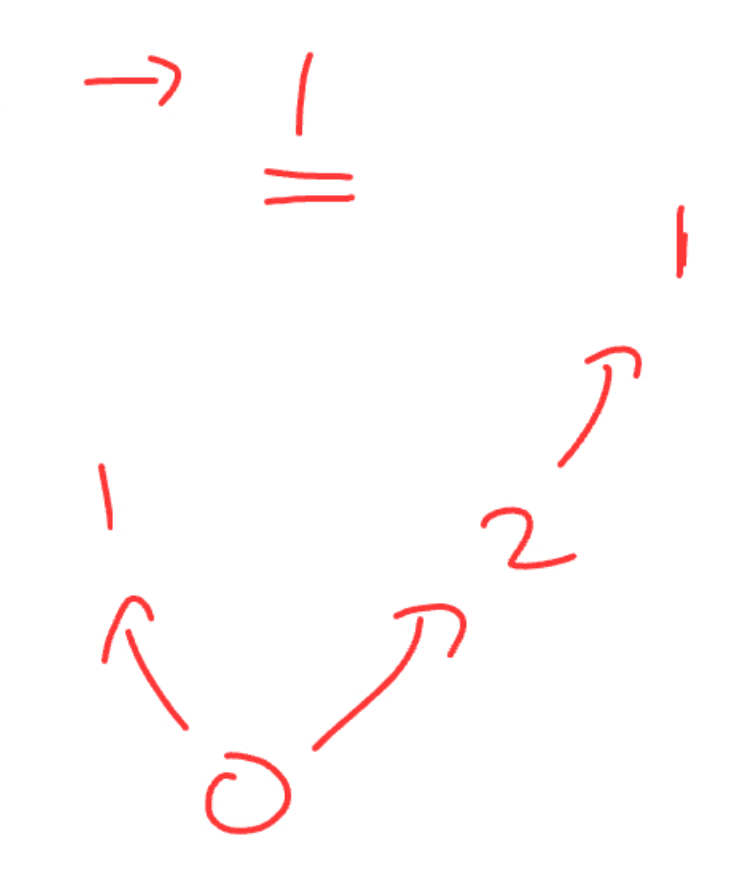
Cyclic.



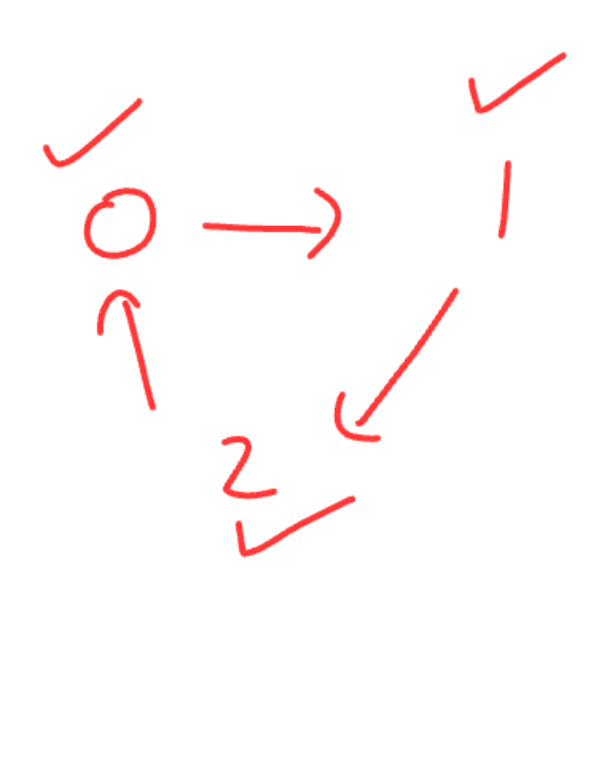
0 → 2
1 → 0, 2
2 →



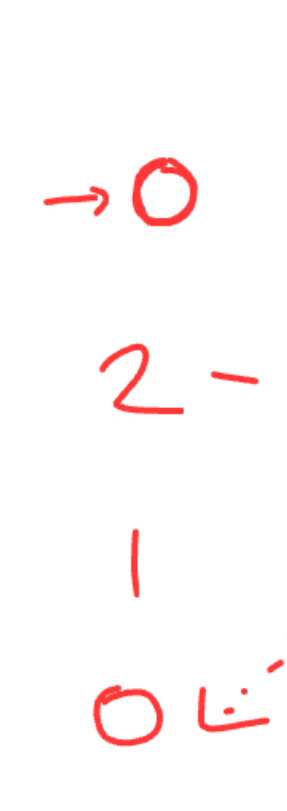
0 → 1, 2
1 →
2 → 1



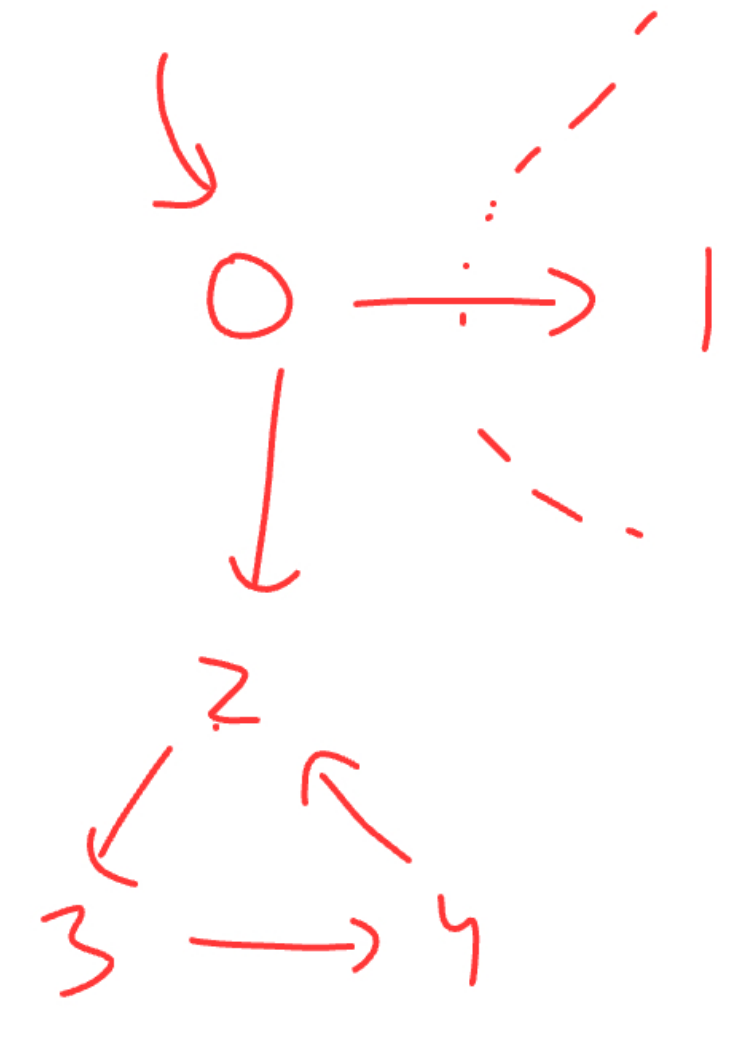
1
2
~~1~~
0

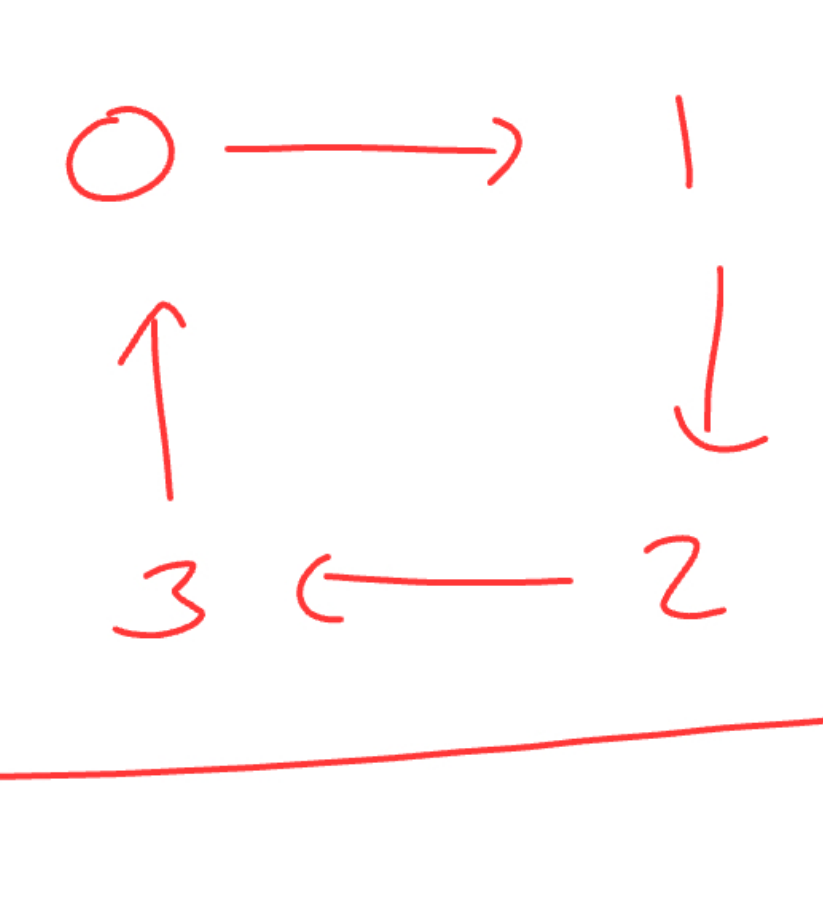


0 → 1
1 → 2
2 → 0



0	1	2
T	T	T

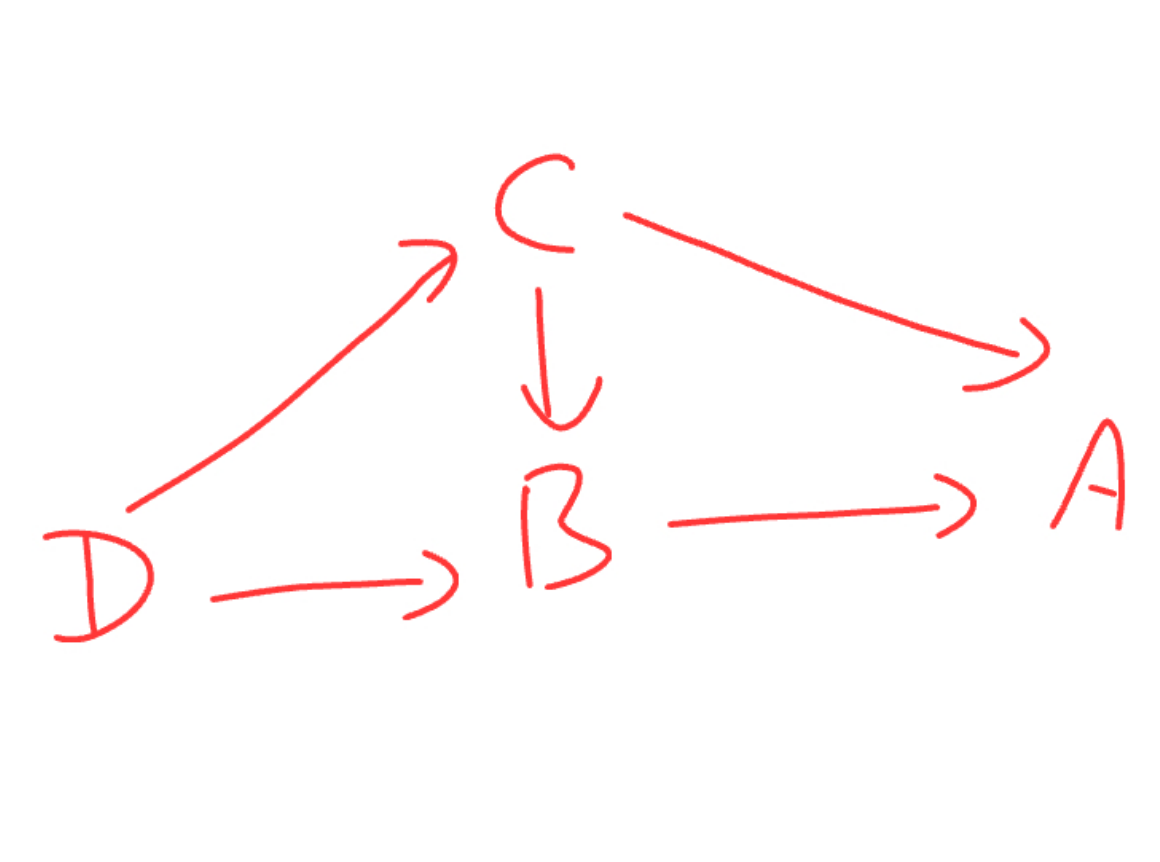




0 1 2 3

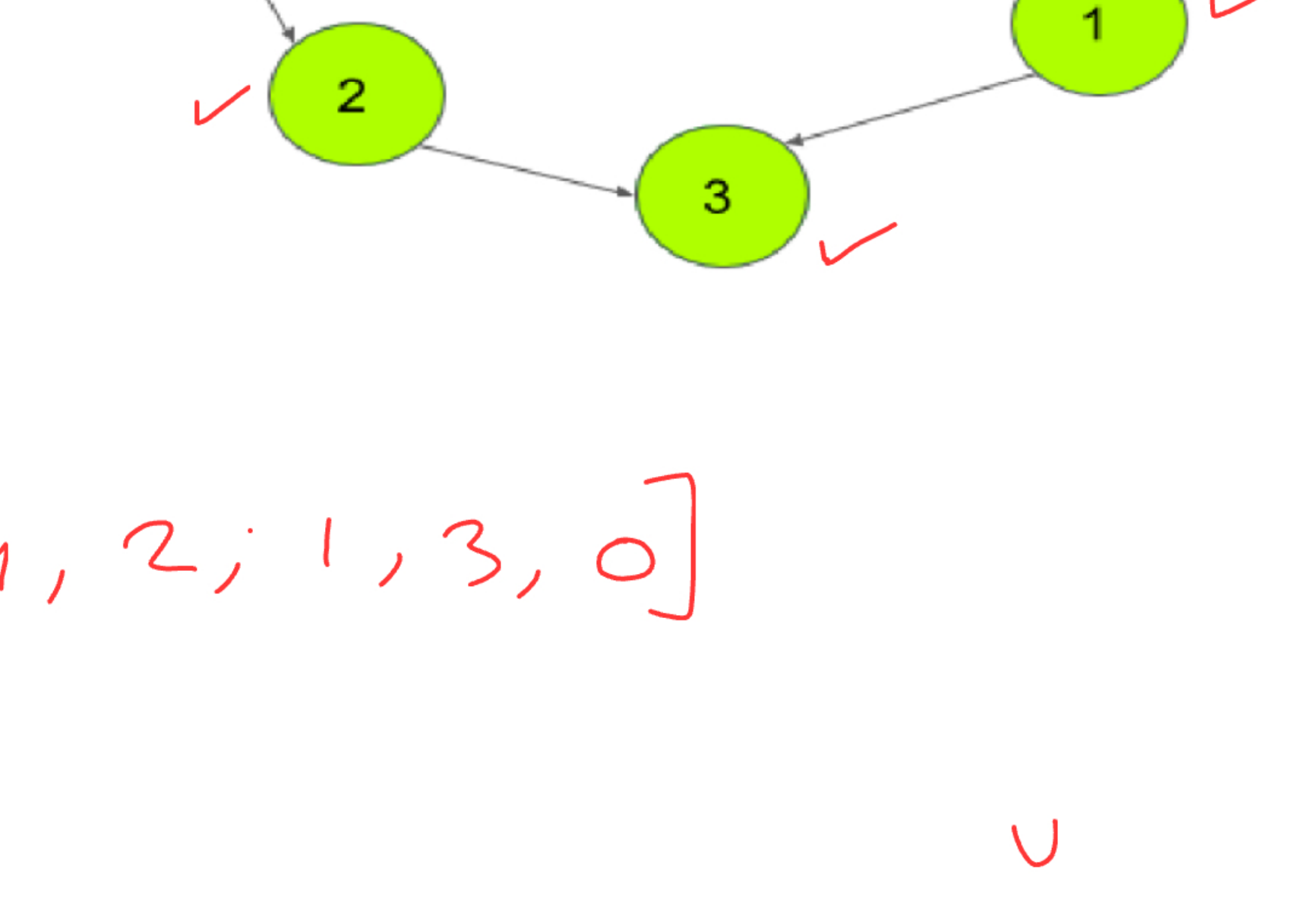
Topological sort.
u → v] → u should come before v.

Course.	Prereqs.
A →	B, C
B →	C, D
C →	D
D →	



D C B A

0 →
1 → 3
2 → 3
3 →
4 → 0, 1
5 → 0, 2



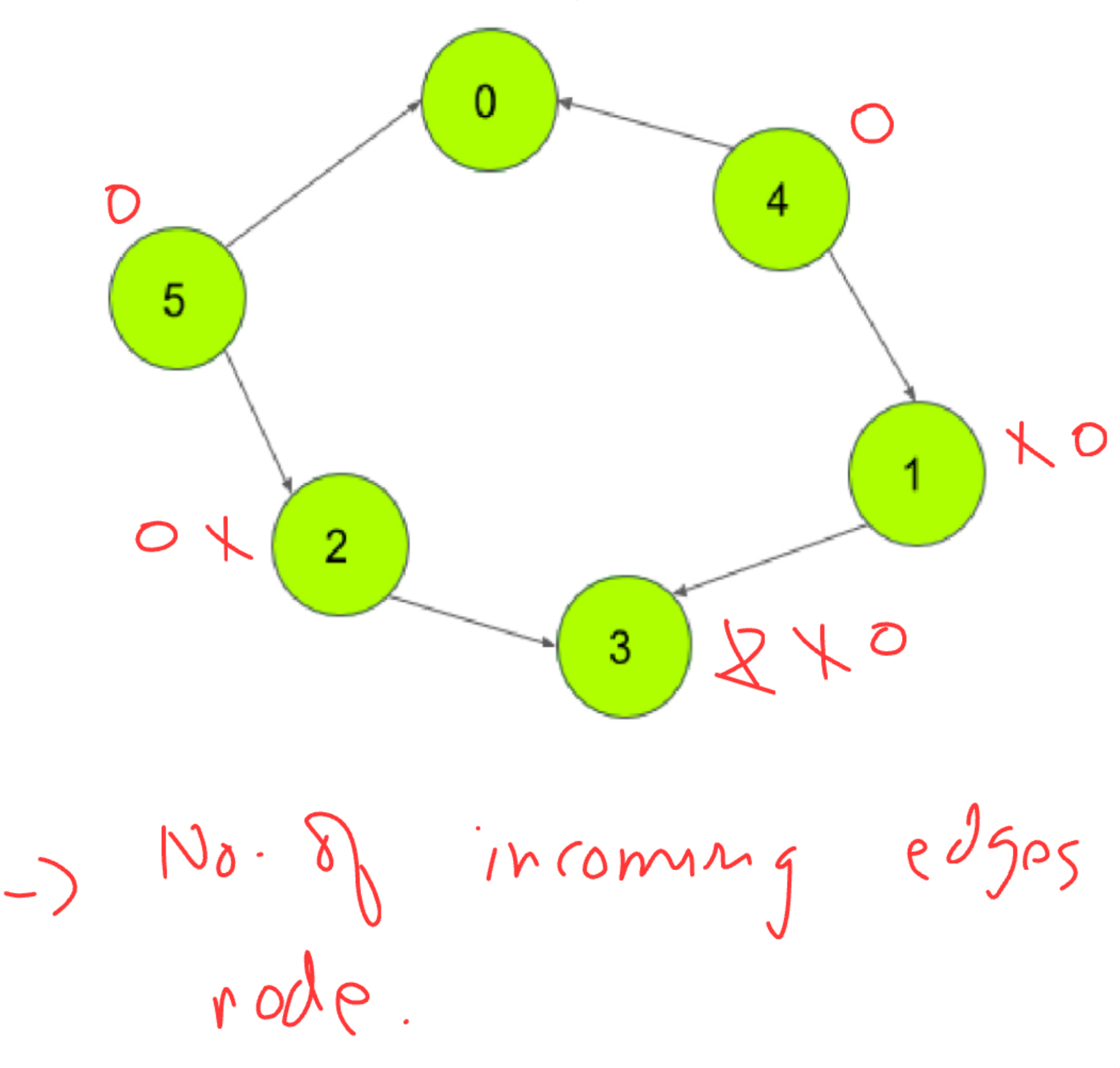
~~5~~
~~4~~
~~2~~
~~1~~
~~3~~
~~0~~

[5, 4, 2, 1, 3, 0]

u

neighbors of v

Indegrees.
0 →
1 → 3
2 → 3
3 →
4 → 0, 1
5 → 0, 2


2 X 0
0
0
4
5
1
X 0
0 X
2
3
2 X 0

Indegree → No. of incoming edges to a node.

Outdegree → No. of outgoing edges from a node.

Q: ~~4~~, ~~5~~, ~~1~~, ~~2~~, ~~0~~, ~~3~~

Ans: [4, 5, 1, 2, 0, 3]